Tutorial No : 5

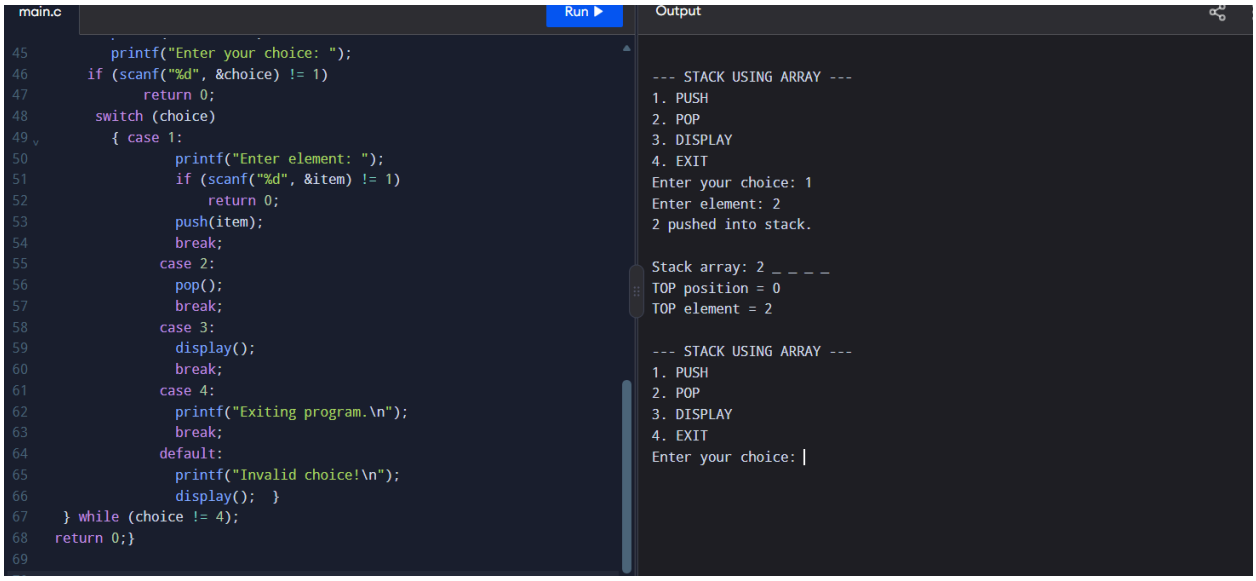
Tutorial Title: Implement Stack and Queue using arrays.
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Program : Stack Using Array

```
#include <stdio.h>
#define MAX 5
int stack[MAX];
int top = -1;
void display()
{  int i;
 printf("\nStack array: ");
if (top == -1)
    {  printf("EMPTY\n");
       printf("TOP = -1\n");
       return; }
 for (i = 0; i < MAX; i++)
    {  if (i <= top)
          printf("%d ", stack[i]);
       else
          printf("_ ");  }
     printf("\nTOP position = %d", top);
    printf("\nTOP element = %d\n", stack[top]);}
    void push(int item)
{ if (top == MAX - 1)
    { printf("Stack Overflow!\n");
       display();
       return; }
     top++;
    stack[top] = item;
    printf("%d pushed into stack.\n", item);
    display(); }
    void pop()
{ if (top == -1)
    {  printf("Stack Underflow! Stack is empty.\n");
```

```
        display();
        return;  }
   printf("Popped element: %d\n", stack[top]);
    top--;
display(); }
int main()
{  int choice, item;
do
    { printf("\n--- STACK USING ARRAY ---\n");
        printf("1. PUSH\n");
        printf("2. POP\n");
        printf("3. DISPLAY\n");
        printf("4. EXIT\n");
        printf("Enter your choice: ");
     if (scanf("%d", &choice) != 1)
            return 0;
      switch (choice)
        { case 1:
              printf("Enter element: ");
              if (scanf("%d", &item) != 1)
                  return 0;
              push(item);
              break;
            case 2:
              pop();
              break;
            case 3:
              display();
              break;
            case 4:
              printf("Exiting program.\n");
              break;
            default:
              printf("Invalid choice!\n");
              display();  }
  } while (choice != 4);
 return 0;}
```

Output:



Program : Queue Using Array

```
#include <stdio.h>
#define SIZE 5
int queue[SIZE];
int front = -1, rear = -1;
void insert() {
   int value;
  if (rear == SIZE - 1) {
      printf("Queue is full\n");
   } else {
      printf("Enter value: ");
      scanf("%d", &value);
     if (front == -1)
         front = 0;
        rear++;
      queue[rear] = value;
   }
}
void delete() {
   if (front == -1 || front > rear) {
      printf("Queue is empty\n");
   } else {
      printf("Deleted: %d\n", queue[front]);
      front++;
      if (front > rear)
         front = rear = -1;    } }
```

```
void display() {
    int i;
   if (front == -1) {
        printf("Queue is empty\n");
    } else {
        for (i = front; i <= rear; i++)
            printf("%d ", queue[i]);
     printf("\n");    } }
int main() {
    int choice;
    do {
        printf("\n1. Insert\n2. Delete\n3. Display\n4. Exit\n");
        printf("Enter choice: ");
      scanf("%d", &choice);
       switch (choice) {
            case 1: insert(); break;
            case 2: delete(); break;
            case 3: display(); break;
            case 4: break;
            default: printf("Invalid choice\n");       }
    } while (choice != 4);
    return 0; }
```

Output:

main.c

Run ▶

```c

#include <stdio.h>
#define SIZE 5

int queue[SIZE];
int front = -1, rear = -1;

void insert() {
    int value;

    if (rear == SIZE - 1) {
        printf("Queue is full\n");
    } else {
        printf("Enter value: ");
        scanf("%d", &value);

        if (front == -1)
            front = 0;

        rear++;
        queue[rear] = value;
    }
}

```

Previous code was kept Restore previous code

Output

```
1. Insert
2. Delete
3. Display
4. Exit
Enter choice: 1
Enter value: 3

1. Insert
2. Delete
3. Display
4. Exit
Enter choice: 1
Enter value: 2

1. Insert
2. Delete
3. Display
4. Exit
Enter choice: 3
3 2

1. Insert
2. Delete
```